

BRENDAN T. CRABB, MD

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EDUCATION

Doctor of Medicine (M.D.) University of Utah School of Medicine, Salt Lake City, UT	May 2023
Bachelor of Science in Physics (B.S.) Point Loma Nazarene University, San Diego, CA	May 2017

POSTGRADUATE TRAINING

Residency, Diagnostic Radiology / T32 Clinician-Scientist Fellow University of California San Diego, La Jolla, CA	2024 – Present
Internship, Internal Medicine National Jewish Health Saint Joseph Hospital, Denver, CO	2023 – 2024
Research Fellowship, Sarnoff Cardiovascular Research Foundation University of California San Diego, La Jolla, CA	2021 – 2022

AWARDS, HONORS, GRANTS

• AHA Postdoctoral Fellowship Award. \$153,000 research grant.	2025 – Present
• JCMR Pohost Prize Finalist , <i>Journal of Cardiovascular Magnetic Resonance</i> , selected by the editorial board as a finalist for the best manuscript of the year.	2024
• JVIR Outstanding Laboratory Investigation Award , <i>Journal of Vascular and Interventional Radiology</i> , one of 5 papers selected by the editorial board for the highest award given to individual papers.	2024
• Young Investigator Award, 1st Prize. North American Society for Cardiovascular Imaging.	2022
• James B. Bassingthwaighe Best Poster Award. International Cardiac Physiome Workshop.	2021
• Sarnoff Cardiovascular Research Foundation Fellowship. \$42,000 research grant.	2021
• AQA Carolyn L. Kuckein Student Research Fellowship Award. \$6,000 research grant.	2020
• RSNA Research and Education Foundation's Medical Student Research Grant. \$6,000 research grant.	2019
• NIH Ruth L. Kirschstein National Research Service Award. \$5,000 research grant.	2019
• PLNU Outstanding STEM Senior Award. Faculty-voted top STEM senior.	2017
• PLNU Valedictorian. Highest academic achievement in graduating class.	2017
• Graduate Summa Cum Laude. Highest academic distinction.	2017

RESEARCH EXPERIENCE

Dept. of Radiology, University of California San Diego, La Jolla, CA
T32 Clinician-Scientist Fellow / AHA Postdoctoral Fellow

Jul. 2024 – Present

Advisor: Albert Hsiao, M.D, Ph.D.

Biventricular Analysis of Regional Myocardial Function in Repaired Tetralogy of Fallot Using a Deep Learning Approach

- Developed deep learning-based approach to quantify regional strain and dyssynchrony directly from cardiac cine SSFP MRI images.
- This project served as the basis for my American Heart Association (AHA) Postdoctoral Fellowship proposal, which was fully funded for \$153,000 over two years.

4D Flow Hemodynamics as an Indicator of RV Dysfunction in Patients with Transposition of the Great Arteries

- Discovered quantitative markers of RV dysfunction using 4D flow ventricular waveforms.
- Aspects of this work have been accepted for oral presentations at the ARRS and SCMR Annual Meetings.

Dept. of Radiology and Bioengineering, University of California San Diego, La Jolla, CA
Sarnoff Cardiovascular Research Fellow

Jul. 2021 – Jul. 2022

Advisors: Andrew D. McCulloch, Ph.D. and Albert Hsiao, M.D, Ph.D.

Deep Learning Analysis and Unsupervised Clustering of Left Ventricular Mechanics in Tetralogy of Fallot

- Investigated left ventricular mechanics in a multi-institutional, international cohort of patients with repaired tetralogy of Fallot using a deep learning-based approach.
- Aspects of this work were accepted for oral presentations at the Radiological Society of North America (RSNA) 2022 Annual Meeting and the North American Society for Cardiovascular Imaging (NASCI) 2022 Annual Meeting.
- Resulted in a first-author publication in the *Journal of Cardiovascular Magnetic Resonance*.

Towards Fully Automated Cardiac Statistical Modeling: A Deep-Learning Based MRI Approach

- Addressed critical bottlenecks in the development of biventricular statistical shape models by automating MRI view classification, short-axis slice selection, phase selection, and landmark localization.
- Resulted in a James B. Bassingthwaite Best Poster Award at the 2021 Cardiac Physiome Workshop, a first-author manuscript in *Journal of Cardiovascular Magnetic Resonance*, and was selected as a finalist for the 2024 JCMR Pohost award, which is given to the best manuscript of the year.

Biventricular Statistical Shape Atlas of Unloaded Reference Geometries: A Novel Method to Control for Hemodynamic Variations in End-Diastolic Pressure

- Developed an unsupervised machine learning approach to minimize geometric shape differences secondary to hemodynamic loading in biventricular statistical atlases, providing a more accurate characterization of underlying ventricular structure.
- Resulted in an oral presentation and conference manuscript at the American Society for Mechanical Engineer's 2022 International Mechanical Engineering Congress and Exposition.

Dept. of Radiology and Imaging Sciences, University of Utah School of Medicine
Research Associate

Dec. 2021 – May 2023

Advisor: Megan Mills, M.D.

Evaluating the Impact of Contextual Interference in Human and Machine Learning

- Modeled the impact of contextual interference and interleaved (non-blocked) training schedules in radiology resident curriculums by using multi-task deep neural networks.
- This project resulted in an oral presentation at the 2023 Association of University Radiologists Annual Meeting.

Dept. of Cardiology, University of Utah School of Medicine

March. 2020 – Jul. 2020

Research Associate

Advisors: Rashmee Shah, M.D.

Electronic Medical Record Review of Patients with Symptoms of Coronavirus Disease 2019

- Designed study, performed statistical analysis, and wrote manuscript comparing the performance of fever, cough, and dyspnea-specific ICD-10 codes with medical record review among patients tested for COVID-19.
- Resulted in a first-author manuscript in *JAMA Network Open*.

Dept. of Neurological Surgery, University of Utah School of Medicine

May. 2020 – Jun. 2022

Research Associate

Advisors: William T. Couldwell, M.D., Ph.D. and Michael Karsy, M.D., Ph.D.

Predicting Readmission Following Pituitary Adenoma Resection using Machine Learning

- Built automated python scripts that process NSQIP data, optimize machine learning models using Bayesian parameter optimization, and run a comparative performance and permutation analysis.
- Resulted in an abstract at the 2020 Weill Cornell Medicine Medical Student Neurological Surgery Research Symposium, abstract at the 2021 Congress of Neurological Surgeons Annual Meeting, and a first-author manuscript in *Neurosurgery*.

High-grade Glioma Imaging Volumes and Survival Analysis using Intraoperative MRI

- Conducted a multivariate regression and survival analysis for patients with high-grade gliomas based on intraoperative MRI volumes, tumor genetics, patient demographics, and clinical features.
- Resulted in a manuscript in the *Journal of Neuro-Oncology*.

Dept. of Radiology and Imaging Sciences, University of Utah School of Medicine

Aug. 2018 – Jul. 2024

Research Associate

Advisors: Gabriel Fine, M.D. and Frederic Noo, Ph.D.

Motion Correction in Digital Subtraction Angiography Using Generative Adversarial Networks

- Developed a generative adversarial network (GAN) that reliably eliminates motion artifacts in visceral angiograms.
- Received the Radiological Society of North America (RSNA) Research and Education Foundation's Medical Student Research Grant, the National Institute of Health (NIH) Ruth L. Kirschstein National Research Service Award Institutional Training Grant, and the AQA Carolyn L. Kuckein Student Research Fellowship Award.
- Resulted in an accepted abstract to the 2019 Radiological Society of North America conference, an oral presentation at the 2020 Society of Interventional Radiology conference, a first-author manuscript in the *Journal of Vascular and Interventional Radiology*, and an international patent pending (WO2021113235A1).

Dynamic Frame Rate Adjustments in Digital Subtraction Angiography

- Designed a system to automatically adjust frame rates in digital subtraction angiography in order to limit radiation exposure and maximize image quality.
- This work resulted in a poster presentation at the 2023 Radiological Society of North America conference and a provisional patent application pending.

Space and Naval Warfare Systems Command, United States Navy, San Diego, CA

Sept. 2017 – Jul. 2018

Civilian Research Assistant / Engineering Contractor

Advisors: Josh Harguess, Ph.D., Katie Rainey, Ph.D., and Niels Olson, M.D.

Whole Slide Image Processing for Deep Learning in Digital Pathology

- Wrote a Python module (SlideSeg) that produces image patches and annotation masks from whole slide images for deep learning in digital pathology.
- Made this tool open-source and publicly available at github.com/btcrabb/slideseg. It has been used by hundreds of researchers across the globe—from the NIH to India.

Synthetic Data Generation for Machine Learning in Domains with Sparse Data

- Researched, built, trained, and deployed generative adversarial networks (GANs) to produce photorealistic synthetic imagery for multiple Department of Defense sponsored projects.

Semantic Segmentation of Integrated Circuits in Printed Circuit Boards

- Developed a novel neural network that achieved state-of-the-art results at integrated circuit detection and segmentation.
- Made a custom version of the Caffe deep learning framework that supported the implementation of conditional random fields as a neural network layer.
- Integrated multiple trained neural network models into C++ software applications.

Image Super Resolution Using Convolutional Neural Networks

- Analyzed the performance of the super-resolution convolutional neural network (SRCNN) at single image super-resolution using multiple image quality assessments.

Department of Inorganic Chemistry, Point Loma Nazarene University, San Diego, CA

May 2015 – May 2017

Undergraduate Research Assistant

Advisor: Laurance Beauvais, Ph.D.

Organocatalysis using N-heterocyclic Carbene Containing Metal-Organic Frameworks

- Designed and performed experiments with a variety of metal-organic frameworks in inert conditions.
- Resulted in an accepted manuscript to the journal *Polyhedron*.

Exploratory Coordination Chemistry: Metal Nitride Nanoclusters with Cobalt, Bismuth, and Titanium Species

- Designed new synthetic routes to titanium and bismuth nanoparticles using metal nitride nanoclusters.
- Resulted in a manuscript and oral presentation at the 2017 Point Loma Nazarene University Honors Scholars Conference.

PROFESSIONAL AND TEACHING EXPERIENCE

Diagnostic Radiology Resident Physician

2024 – Present

University of California San Diego, La Jolla, CA

- Resident physician in the UCSD clinician-scientist program, a five-year diagnostic radiology residency with an additional protected year of NIH-supported research.

Internal Medicine Intern

2023 – 2024

National Jewish Health | Saint Joseph Hospital, Denver, CO

- Preliminary internship in internal medicine.

Sarnoff Cardiovascular Research Fellow

2021 – 2022

University of California San Diego, La Jolla, CA

- Fully funded, one year research fellowship at an institution of my choosing through the Sarnoff Cardiovascular Research Foundation, designed to train the next generation of leaders in cardiovascular innovation, research, and medicine.

- Selected to study under the co-mentorship of Dr. Albert Hsiao, associate professor of Radiology, and Dr. Andrew McCulloch, the Shu Chien Chancellor's Endowed Chair in Engineering and Medicine and director of the Institute for Engineering in Medicine, at the University of California San Diego.

Course Instructor

2017 – 2018

Eduonix Learning Solutions

- Designed and developed multiple online courses on machine learning, including recorded lectures, programming projects, and quizzes. To date, over 25,000 students worldwide have enrolled in these courses.
- Courses included "Machine Learning for Absolute Beginners", "Projects in Machine Learning: Beginner to Professional", "Mathematical Foundations for Machine Learning and AI", and "Applied Machine Learning for Healthcare".

Machine Learning Engineering Contractor

2017 – 2018

Space and Naval Warfare Systems Command, U.S. Navy, San Diego, CA

- Full time machine learning engineering contractor for the U.S. Navy, hired due to my performance as an intern through the Naval Research Enterprise Internship Program.
- Tasked with researching and applying deep learning, including image super-resolution, generative adversarial networks, and semantic segmentation, to several Department of Defense sponsored projects.

Naval Research Enterprise Intern

2017

Office of Naval Research, U.S. Navy, San Diego, CA

- Summer internship studying Naval applications of computer vision and artificial intelligence.
- Personally recruited due to my performance in my undergraduate mathematics courses.

Chemistry Research Assistant

2015 – 2017

Point Loma Nazarene University, San Diego, CA

- Undergraduate research under Dr. Laurance Beauvais in inorganic chemistry.

University Physics I and II Teaching Assistant

2015 – 2017

Point Loma Nazarene University, San Diego, CA

- Facilitated lab experiences and graded problem sets for several undergraduate physics courses.

Math and Physics Tutor

2013 – 2017

Point Loma Nazarene University, San Diego, CA

- Tutored and assisted individuals and groups of students over a broad range of math and physics concepts through the Math Learning Center.

COMMUNITY INVOLVEMENT

Medical Volunteer

Mar. 2023 – Apr. 2023

RAD-AID international

- Partnered with RAD-AID, PEARLS (POCUS for Emergency and Acute care in Resource-Limited Settings), and the University of Utah to complete a four-week volunteer assignment in Vanuatu, an underserved island nation in the South Pacific.
- Taught multiple point-of-care ultrasound workshops to local Vanuatu physicians and trainees.
- Completed the RAD-AID Radiology Readiness and Tele-ultrasound surveys for the Vila Central Hospital.

Co-President	2019 – 2021
Neurosurgery Student Interest Group	
Organized lunch talks with faculty and host clinical skills workshops with the neurosurgery residents.	
Medical Volunteer	2019 – 2020
Junior League Community Assistance and Resource Event (CARE) Fair, Salt Lake City, UT	
Obtained histories and performed physical exams at the CARE Fair, which is an annual two-day event that provides underserved members of the community with routine medical, dental, and vision services.	
Medical Volunteer	2018 – 2019
Midvale Community Clinic, Salt Lake City, UT	
Obtained histories and performed physical exams at the Midvale student-run clinic.	
Mentor and Tutor	2015 – 2017
Reality Changers Tutoring Ministries, City Heights San Diego, CA	
Mentored and tutored underserved high school students in the inner city of San Diego on a weekly basis.	

PEER REVIEWED JOURNAL ARTICLES

- Crabb, B. T.**, Chandrupatla, R., Masutani, E., et al. "Characteristics of Left Ventricular Dysfunction in Repaired Tetralogy of Fallot: A Multi-Institutional Deep Learning Analysis of Regional Strain and Dyssynchrony." *Journal of Cardiovascular Magnetic Resonance* (2025): 101886; <https://doi.org/10.1016/j.jocmr.2025.101886>. 2025
- Govil, S.*, **Crabb, B. T.***, Deng, Y., et al. "A Deep Learning Approach for Fully Automated Cardiac Shape Modeling in Tetralogy of Fallot." *Journal of Cardiovascular Magnetic Resonance* 25.1 (2023): 15. * Contributed equally as co-first authors. 2023
- Gamboa, N. T., **Crabb, B. T.**, Henson, J. C., et al. "High-grade glioma imaging volumes and survival: A single-institution analysis of 101 patients after resection using intraoperative MRI." *Journal of Neuro-Oncology* (2022): 1-11. 2022
- Crabb, B. T.**, Hamrick, F., Richards, T., et al. "Deep Learning Subtraction Angiography: Improved Generalizability with Transfer Learning." *Journal of Vascular and Interventional Radiology*. 34.3 (2022): 409-419. 2022
- Crabb B. T.**, Govil, S., Hegde, S., et al. "Biventricular Statistical Shape Atlas of Unloaded Reference Geometries: A Novel Method to Control for Hemodynamic Variations in End-Diastolic Pressure." *ASME International Mechanical Engineering Congress and Exposition*. 86717 (2022). 2022
- Crabb, B. T.**, Hamrick, F., Campbell, J., et al. "Machine Learning-Based Analysis and Prediction of Unplanned 30-Day Readmissions after Pituitary Adenoma Resection: A Multi-Institutional Retrospective Study with External Validation." *Neurosurgery* (2022): 10-1227; doi: 10.1227/neu.0000000000001967 2022
- Crabb, B. T.**, Lyons A., Bale M., et al. "Comparison of International Classification of Diseases and Related Health Problems, Tenth Revision Codes with Electronic Medical Records Among Patients with Symptoms of Coronavirus Disease 2019." *JAMA Network Open*. 2020;3(8). doi:10.1001/jamanetworkopen.2020.17703 2020
- Crabb, B. T.**, Olson, N. H. "SlideSeg: A Python module for the creation of annotated image repositories from whole slide images", *Proc. SPIE, Medical Imaging 2018: Digital Pathology*, Volume 10581, 12 Feb. 2018. 2018

9. Ward, C. M., Harguess, J., **Crabb, B. T.**, Parameswaran, S. "Image quality assessment for determining efficacy and limitations of Super-Resolution Convolutional Neural Network (SRCNN)", *Proc. SPIE 10396, Applications of Digital Image Processing XL, 1039605*, 19 September 2017; doi: 10.1117/12.2275157 2017
10. Schumacher, W. T., Matthews, M. J. Larson, S. A., Lemmon, C. E., Campbell, K. A., **Crabb, B. T.**, Chicoine, B. J., Beauvais, L. G., Perry, M. C. "Organocatalysis by site-isolated N-heterocyclic carbenes doped into the UIO-67 framework." *Polyhedron*, Volume 114, 16 August 2016, Pages 422-427 2016

PEER REVIEWED ABSTRACTS

1. **Crabb, B. T.**, Wong, S. Y., Rushworth, P., Alshawabkeh, L., Valdez-Jasso, D., Hsiao, A. "Feasibility of 4D Flow Hemodynamics as an Indicator of RV Dysfunction in Patients with Transposition of the Great Arteries." *Journal of Cardiovascular Magnetic Resonance* 27 (2025). 2025
2. Tandon, A., Deng, L., Nasopoulou, A., **Crabb, B. T.**, Young, A., Hussain, T., Karamlou, T., Zahka, K., Nguyen, C. "Adverse Biventricular Shape and Exercise Capacity in Repaired Tetralogy of Fallot." *Journal of Cardiovascular Magnetic Resonance* 26 (2024). 2024
3. **Crabb, B. T.**, Noo, F., and Fine, G. C. "Image synthesis for motion correction in digital subtraction angiography using a generative adversarial network (GAN)." *Journal of Vascular and Interventional Radiology* 31.3 (2020): S135. 2020
4. Schumacher, W., Mathews, M., Larson, S., Lemmon, C., Campbell, K., **Crabb, B. T.**, Chicoine, B., Beauvais, L., Perry, M. "Synthesis and characterization of MOF-supported NHC catalysts" In *Abstracts of Papers of the American Chemical Society* 251 (2016). 2016

ORAL PRESENTATIONS

1. **Crabb, B. T.**, Wong, S., Rushworth, P, et al. (2025, April). "Feasibility of 4D Flow Hemodynamics as an Indicator of RV Dysfunction in Patients with Transposition of the Great Arteries." *American Roentgen Ray Society 2025 Annual Meeting*, San Diego, CA (Accepted) 2025
2. **Crabb, B. T.** (2025, Feb). "A deep learning approach for fully automated cardiac shape modeling in tetralogy of Fallot." *Caring for Adults and Teens with Congenital Heart Disease (CATCH)*, Oahu, HI (Invited Speaker). 2025
3. **Crabb, B. T.**, Wong, S., Rushworth, P, et al. (2024, Jan). "Feasibility of 4D Flow Hemodynamics as an Indicator of RV Dysfunction in Patients with Transposition of the Great Arteries." *Society for Cardiovascular Magnetic Resonance 28th Annual Scientific Sessions*, Washington, DC 2025
4. **Crabb, B. T.** (2024, March). "Trials, Abstracts, and What It Takes to Publish an Award-Winning Article: Deep Learning Subtraction Angiography." *Society of Interventional Radiology 2024 Annual Meeting*, Salt Lake City, UT (Invited Speaker). 2024
5. Tandon, A., Deng, L., Nasopoulou, A., **Crabb, B. T.**, et al. (2024, Jan). "Adverse Biventricular Shape and Exercise Capacity in repaired Tetralogy of Fallot." *Society for Cardiovascular Magnetic Resonance 2024 Annual Meeting*, London, UK. 2024
6. **Crabb, B. T.**, Hartwell, C., Mills, M. (2023, April). "Learning from Machine Learning: Evaluating How Contextual Interference Impacts the Learning of a Radiology Task in Machines and Radiologists." *Association of University Radiologists 2023 Annual Meeting*, Austin, TX. 2023

7. **Crabb, B. T.**, Chandrupatla, R., Masutani, E., et al. (2022, Nov). "Deep Learning Left Ventricular Mechanical Analysis – Sensing Bi-Ventricular Dysfunction in Tetralogy of Fallot" *Radiological Society of North America 2022 Annual Scientific Meeting*, Chicago, IL. 2022
8. Chandrupatla, R., **Crabb, B. T.**, Masutani, E., et al. (2022, Nov). "Deep Learning Visualization and Quantification of Mechanical Dyssynchrony in Cardiac MRI" *Radiological Society of North America 2022 Annual Scientific Meeting*, Chicago, IL.
9. **Crabb, B. T.**, Chandrupatla, R., Masutani, E., et al. (2022, Sept). "Deep Learning Analysis and Unsupervised Clustering of Left Ventricular Mechanics in Tetralogy of Fallot" *North American Society for Cardiovascular Imaging 2022 Annual Scientific Meeting*, Cleveland, OH. 2022
10. **Crabb B. T.** (2022, June). Quantitative Analysis of Cardiac MRI: From Deep Learning Synthetic Strain to Bi-Ventricular Computational Modeling. Oral presentation at the *42nd annual Sarnoff Cardiovascular Research Foundation Scientific Meeting*, Arlington, VA. 2022
11. **Crabb B. T.**, Hamrick, F., Campbell, J., et al. (2021, Oct). Machine Learning-Based Analysis and Prediction of Unplanned 30-Day Readmissions after Pituitary Adenoma Resection: A Multi-Institutional Retrospective Study with External Validation. Oral presentation at the *2021 Congress of Neurological Surgeons Annual Meeting*, Austin, TX. 2021
12. **Crabb, B. T.**, Noo, F., Fine, G. C., (2020, April). Image Synthesis for Motion Correction in Digital Subtraction Angiography (DSA) Using a Generative Adversarial Network (GAN). Oral presentation at the *Society of Interventional Radiology's 2020 Annual Scientific Meeting*. 2020
13. **Crabb, B. T.**, Hamrick, F., Campbell, J. et al. (2020, Jan) Predicting Readmission Following Pituitary Adenoma Resection: A Machine Learning Approach. Oral presentation at the *Weill Cornell Medicine Medical Student Neurological Surgery Research Symposium*. 2020
14. **Crabb, B. T.**, Olpin, J., Fine, G. C., (2019, Sept). A Case of Budd-Chiari Syndrome. Clinical case presentation at the *Imaging Elevated: Utah Symposium for Emerging Investigators*, Salt Lake City, UT. 2019
15. **Crabb, B. T.**, Shoukry, M., Beauvais, L., Bennett, M. V., (2017, May). Exploratory Coordination Chemistry: Metal Nitride Nanoclusters with Cobalt, Bismuth, and Titanium Species. Oral presentation at the *2017 Point Loma Nazarene University Honors Scholars Conference*, San Diego, CA. 2017

POSTER PRESENTATIONS

1. **Crabb, B. T.**, Noo, F., Fine, GC. (2023, Dec). Fully Automated Dynamic Frame Rate Adjustment in Digital Subtraction Angiography. Poster presented at the *2023 Radiological Society of North America Annual Scientific Meeting*, Chicago, IL. 2023
2. **Crabb, B. T.**, Chandrupatla, R., Masutani, E.M., et al. (2022, Apr). Deep Learning Left Ventricular Mechanical Analysis – Sensing Bi-Ventricular Dysfunction in Tetralogy of Fallot. Poster presented at the *Rady Children's Hospital San Diego Pediatrics Research Symposium*. San Diego, CA. 2022
3. **Crabb, B. T.**, Govil, S., Young, A.A., et al. (2021, Nov). Towards Fully Automated Cardiac Statistical Shape Modeling: A Deep-Learning Based MRI View and Phase Selection Tool. Poster presented at the *2021 Cardiac Physiome Workshop*. 2021
4. **Crabb, B. T.**, Noo, F., and Fine, G. C. (2021, Nov). Motion Correction in Digital Subtraction Angiography using Generative Adversarial Networks: An Implementation and Evaluation of the Gradient-Consistency Loss Function. Poster presented at the *2021 Radiological Society of North America Annual Scientific Meeting*, Chicago, IL. 2021
5. Cogman, A., Eastaway, A., Fine, C., Smith, T., Weintraub, M., Kaufman, C., Quencer, K., **Crabb, B. T.**, Quigley, E. (2019, Dec). Creation of a Haptic 3D Printed Simulator for TIPS Training in Augmented and

Virtual Reality. Poster presentation at the Radiological Society of North America's 2019 Annual Scientific Meeting, Chicago, IL.	2019
6. Crabb, B. T. , Fine, G. C., (2019, Dec). Motion Correction in Digital Subtraction Angiography Using Machine Learning. Poster presented at the Radiological Society of North America's 2019 Annual Scientific Meeting, Chicago, IL.	2019
7. Crabb, B. T. , Olson, N. H., (2018, Feb). SlideSeg: A Python module for the creation of annotated image repositories from whole slide images, Poster presented at SPIE, Medical Imaging 2018: Digital Pathology.	2018
8. Crabb, B. T. , Ward, C., (2018, Feb). Improving Semantic Segmentation Results using a CRF-RNN. Poster presented at the SPAWAR Naval Applications of Machine Learning Workshop, San Diego, CA.	2018
9. Crabb, B. T. , Campbell, K., (2017, April). Preparation of metal-organic frameworks containing N-heterocyclic carbene catalysts. Poster at the 253rd ACS National Meeting and Exposition, San Francisco, CA	2017

ONLINE PUBLICATIONS

1. **Crabb, B. T.**, Karsy, M., Roper, S., Still, M., Vega, R. (2022, July). "Machine Learning-based Analysis and Prediction of Unplanned 30-Day Readmissions", Congress of Neurological Surgeons Journal Club Podcast. <https://www.cns.org/publications/journal-club-podcasts/podcast-detail/august-2022-machine-learning-based-analysis-predic> 2022

PATENTS

1. Crabb, B.T., Noo, F., Fine, G.C. Medical Image Synthesis for Motion Correction Using Generative Adversarial Networks. International Patent Application No. WO 2021/113235 A1, filed December 02, 2019, published June 10, 2021.
2. Crabb, B.T., Noo, F., Fine, G.C. Dynamic Frame Rate Adjustments for Digital Subtraction Angiography. International Patent Application Serial No. 63/464,286, filed on May 5, 2023.

MANUSCRIPT REVIEWS

Ad hoc peer reviewer for manuscripts and proposals submitted to:

- Journal of Imaging Informatics in Medicine 2023 – Present
- Mitacs Accelerate 2022
- IEEE Journal of Biomedical and Health Informatics 2021

PROFESSIONAL AFFILIATIONS AND CERTIFICATIONS

American Roentgen Ray Society (ARRS)	2024 – Present
Advanced Cardiovascular Life Support (ACLS) certification	2023 – Present
American Heart Association (AHA)	2021 – Present
Advanced Wilderness Life Support (AWLS) certification	2020 – Present
Society of Interventional Radiology (SIR)	2018 – Present
Radiological Society of North America (RSNA)	2018 – Present
Secret-level security clearance, United States Government, active	2017 – Present